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TriGWONet a lightweight multibranch convolutional neural network using gray wolf optimization for accurate oral cancer image classification

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Abstract

Oral cancer remains a critical global health burden, with survival rates heavily dependent on early and accurate diagnosis. In recent years, deep learning-based medical image analysis has emerged as a promising tool for automated detection; however, existing models often suffer from excessive architectural complexity, limited accuracy, and inadequate generalization to clinical data. To address these limitations, we propose a novel lightweight multi-branch convolutional neural network (CNN) architecture integrated with Transformer-inspired design principles and optimized through Gray Wolf Optimization (GWO). The model comprises three parallel feature extraction branches, each incorporating SeparableConv2D layers, Batch Normalization, ReLU activation, and Global Max Pooling, followed by dense layers for hierarchical representation learning. GWO is employed to dynamically optimize critical hyperparameters—including the number of filters, dense layer neurons, and dropout rates—resulting in a compact and highly performant configuration. Performance is assessed on the OCI dataset (240 images; lips and tongue; cancerous vs. non-cancerous) with matched input size, augmentations, and tuning procedures for contemporary lightweight baselines (MobileViT, ConvNeXt, CoAtNet). The proposed TriGWONet model achieved a classification accuracy of 99.64%, sensitivity of 99.60%, specificity of 99.70%, precision of 99.50%, and an ROC AUC of 1.00. Comparative analysis demonstrates that our method significantly outperforms state-of-the-art models, including MobileViT, ConvNeXt, and CoAtNet. The superior accuracy and minimal computational footprint of the proposed TriGWONet model underscore its potential for real-time clinical deployment and pave the way for broader applications in medical image-based disease classification.

Keywords Oral cancer detection, Lightweight deep learning, MobileViT transformer, Squeeze-and-Excitation (SE) module, Medical imaging

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1 Introduction

Oral cancer constitutes a major public health concern globally, accounting for a significant proportion of head and neck malignancies. Its prevalence is particularly high in low- and middle-income countries such as India and Bangladesh, where risk factors like tobacco use, betel quid chewing, and limited access to routine screening exacerbate the burden. Despite advancements in treatment modalities, the five-year survival rate for oral cancer remains low, primarily due to late-stage diagnosis. Early detection can drastically improve treatment outcomes and reduce mortality.

In this context, artificial intelligence (AI), and more specifically deep learning, has emerged as a transformative force in medical diagnostics, offering the ability to analyze complex visual patterns and support clinical decision-making with high accuracy. Recent years have witnessed the proliferation of deep learning models, particularly convolutional neural networks (CNNs), in the domain of medical image classification. While CNN-based models such as ResNet, VGGNet, and DenseNet have demonstrated commendable success, their performance often deteriorates when applied to limited and noisy medical datasets.

Transformer-based vision models capture long-range dependencies and often excel when trained on large, diverse datasets with substantial computational and memory budgets. In small-data, resource-constrained clinical settings, however, they typically require heavy regularization or pretraining and incur higher inference latency and memory footprint. Compact CNNs, especially those employing depthwise-separable convolutions and non-parametric pooling, preserve useful inductive biases such as locality and translation equivariance, yielding more favorable accuracy–efficiency trade-offs. Given the limited labeled data, heterogeneous acquisition conditions, and the need for reliable, fast inference on modest hardware, a carefully engineered lightweight CNN is a more suitable operating point than introducing transformer components.Transformer-based architectures like Vision Transformers (ViT), MobileViT, and CoAtNet have also been introduced for their capacity to model long-range dependencies. However, these models are frequently characterized by high computational demands, excessive parameter counts, and a tendency to overfit, especially in scenarios involving small-scale datasets. Moreover, most existing frameworks lack adaptive tuning mechanisms and are not optimized for real-time clinical deployment due to their complexity and resource requirements.These limitations underscore the urgent need for a lightweight, high-performing model capable of generalizing well on medical image data while maintaining computational efficiency. A model that integrates multiple feature extraction pathways, utilizes parameter-efficient convolution techniques like SeparableConv2D, and leverages non-parametric pooling strategies such as Global Max Pooling can offer a balanced trade-off between accuracy and speed. Furthermore, employing a metaheuristic algorithm like gray wolf optimization (GWO) for hyperparameter tuning can significantly enhance model adaptability and reduce manual intervention, especially in tuning dropout rates, convolutional filter sizes, and dense layer configurations.To address these challenges, we propose a lightweight, GWO-optimized multi-branch CNN-Transformer hybrid model tailored for oral cancer image classification. TriGWONet is distinguished from prior lightweight approaches in both architecture and optimization. Architecturally, it adopts a three-branch, multi-scale design: parallel streams with heterogeneous receptive fields (for example, 3×3, 5×5, and 7×7) each consist of SeparableConv2D, Batch

Normalization, ReLU, and Global Max Pooling, after which the branch embeddings are fused by a minimal classification head. This configuration broadens representational coverage without the depth or width expansion typical of single-path backbones. On the optimization side, the network is coupled with gray wolf optimization (GWO) to jointly select branch filter widths, dense-layer sizes, dropout, learning rate, and batch size under a fixed search budget. Whereas many lightweight models rely on manual heuristics or tune only a subset of hyperparameters, this population-based, end-to-end search yields compact configurations that remain competitive under matched training conditions, clarifying the combined contribution of the tri-branch design and optimizer-guided tuning.

2 Related work

Song et al. [1] explored model credibility in medical image analysis by using a Bayesian deep learning setup for oral cancer detection. Standard deep neural networks often don't measure uncertainty, which is key in clinical choices. To fix this, the authors used a method that predicts confidence and classification accuracy. They trained their model using images of cheek lining from people at high risk, taken with a special device. The Bayesian setup could point out unclear cases by measuring uncertainty. The study also found that checking cases with uncertainty over 0.3, or the top 10% most uncertain guesses, made classification more reliable, reaching almost 90% accuracy. This way, diagnostic confidence goes up, and it offers a way to prioritize cases for review, making it easier to use in clinics.Jeyaraj and Nadar [2] created a system to help diagnose oral cancer from hyperspectral images, using a special deep CNN. The model uses a partitioning method in the CNN to separate and sort areas in high-dimensional data. Tested on clinical data, the system got 91.4% accuracy, with 0.94 sensitivity and 0.91 specificity, when telling apart bad and harmless lesions. When tested on a larger set of 500 samples, accuracy rose to 94.5%. These results show that deep learning is helpful in hyperspectral analysis for finding oral cancers early.Warin et al. [3] suggested an automatic screening system that puts together image classification and lesion detection to spot oral squamous cell carcinoma. The authors used 700 clinical photos, half with cancer and half healthy, and used DenseNet121 for classification. It did very well, with 99% precision, 100% recall, and 98.75% sensitivity. For finding lesions, they used Faster R-CNN, getting an F1-score of 79.31% and a precision-recall AUC of 0.79. This two-part approach shows that CNN systems can accurately classify and find lesions in real clinical photos.Song et al. [4] examined transformer models, like Vision Transformer (ViT) and Swin Transformer, for classifying mobile oral cancer images. Their work aimed to give a cheap way to diagnose high-risk groups. The Swin Transformer did best, hitting 88.7% accuracy, better than ViT (by 2.3%), VGG19 (85.2%), and ResNet50 (84.5%). The study found that transformers are good at picking up on both broad and specific image details, which makes them suited for mobile screening where it's hard to get tools.Welikala et al. [5] made a deep learning setup as part of MeMoSA® (Mobile Mouth Screening Anywhere) to spot and sort oral lesions. The setup used a tool where doctors across the world could label data. To keep labels alike, they came up with a way to group input from many experts. They used ResNet-101 on this data to classify images and Faster R-CNN to find lesions. ResNet-101 got F1-scores of 87.07% for finding lesions and 78.30% for spotting cases that need a referral. Faster R-CNN got a lower F1-score of 41.18% for spotting

lesions needing referral. These results back up the idea that deep learning can help build cheap diagnosis tools for oral cancer, mostly where people don't have good access to care.Song et al. [6] built a small deep learning model for use on phones to help screen in faraway places. They used data from over 5,000 patients, gathered through special mobile tools. They trained a MobileNet setup with focal loss to deal with uneven data and shrunk the model to a 16.3 MB TensorFlow Lite file for Android phones. Put into an app, the system could sort lesion images in less than 300 milliseconds on a Moto G5, with 81% accuracy in telling apart risky from normal tissue. This shows it's doable to use AI screening tools offline, in real time.Song et al. [7] addressed class imbalance in oral cancer datasets from high-risk groups. They built a dataset of 3,851 cheek mucosa images using a mobile screening tool. To balance classes, they used methods such as weighted loss, data increase, undersampling, focal loss, and ensemble models. This combined approach improved the identification of underrepresented categories, such as premalignancy cases, which is helpful for early cancer detection in communities that have limited resources.Hsieh et al. [8] studied deep learning to find subtle changes in chemoresistant oral cancer cells that are hard to see with standard microscopy. They used EfficientNet-B3, with over- and under-sampling, to classify non-cancerous, non-resistant, and chemoresistant cells with at least 98.33% accuracy. Ensemble learning improved the results when looking at mixed cell populations, reducing RMSE to 0.16 in chemoresistance estimation. These results show that the model can capture small changes, making it useful for drug response analysis in vitro.Deo et al. [9] created an ensemble-based deep learning model to classify oral squamous cell carcinoma from histopathological images, to reduce diagnostic differences among pathologists. The study used 696 annotated biopsy images, processed with a 2D empirical wavelet transform for feature extraction. They used a mix of pretrained ResNet50 and DenseNet201 for classification. The ensemble had a classification accuracy of 92%, with good sensitivity and specificity. This shows that combining signal processing with deep learning can improve diagnostic reliability in pathology.Bakare and Kumarasamy [10] used a standard machine learning method to classify oral cancer in histopathological images. They used median filtering to remove noise, and then extracted features—energy and entropy measures from color channels. They compared Support Vector Machines (SVM) and K-Nearest Neighbors (KNN), and SVM achieved 98% accuracy, which was better than KNN's 83%. These results confirm that SVM is good for histological analysis in oral cancer diagnostics.To deal with differences in smartphone-taken images, Lin et al. [11] suggested a mobile deep learning system for oral cancer screening. The study used a standard image process called the "centered rule" to reduce differences. They tested HRNet on a five-class dataset of 455 images. The results were: 83.0% sensitivity, 96.6% specificity, 84.3% precision, and an F1-score of 83.6%. The "centered rule" improved the F1-score by 8%, and a resampling strategy added another 6%. This shows that standard data collection and architectures can improve performance in mobile AI screening systems.Chakraborty et al. [12] proposed a non-invasive prescreening method using digital infrared thermal imaging (DITI) to detect oral cancer, particularly in underserved populations. Unlike invasive biopsies that risk false negatives, their method used bilateral facial thermal asymmetry as a diagnostic marker. Texture features were extracted from frontal and lateral thermograms, then clustered using k-means and fuzzy k-means. Classification was performed via cluster voting. The best performance—86.12% accuracy—was achieved by fusing

features from both halves of the frontal face, improving their earlier model by 6.5%. This study highlights the potential of DITI as a low-cost, non-invasive screening tool.Aubreville et al. [13] applied deep learning to confocal laser endomicroscopy (CLE) images for detecting oral squamous cell carcinoma. Their CNN model analyzed 7,894 CLE images from various oral regions and was benchmarked against texture-based methods. The proposed framework achieved an AUC of 0.96 and a mean accuracy of 88.3%, with 86.6% sensitivity and 90% specificity, surpassing prior CLE-based techniques. These findings affirm CLE's potential as a real-time, high-resolution diagnostic method when combined with automated deep learning.Figueroa et al. [14] focused on enhancing model interpretability by integrating guided attention into CNNs for oral cancer classification. They developed a dual-stage framework using the Guided Attention Inference Network (GAIN), combining classification, attention mining, and bounding box refinement. Grad-CAM was employed to visualize decision-making. Images were captured using mobile devices and augmented for robustness. The model successfully aligned attention with clinically significant regions, improving both accuracy and transparency—an essential step toward real-world deployment where interpretability is critical.Bansal et al. [15] assessed deep transfer learning setups, enhanced with hybrid methods, for early oral cancer detection using real-time and histopathological images. The team preprocessed ResNet50, MobileNetV2, VGG16, VGG19, and DenseNet models with Gaussian filtering and morphological cropping. After tuning with different methods, DenseNet combined with a hybrid method gave the best result, with 95.41% validation accuracy and low RMSE. This points to the value of mixing transfer learning and tuning for solid oral cancer sorting.Ariji et al. [16] checked out a deep learning system for finding cervical lymph node spread in oral squamous cell carcinoma using contrast-enhanced CT scans. Looking at 127 nodes with spread and 314 without from 45 patients, the model got 78.2% accuracy, 75.4% sensitivity, and 81.0% specificity, with an AUC of 0.80—similar to two expert radiologists' opinions. This hints that the model could work as a support tool in clinics for checking lymph node issues.Ilhan et al. [17] gave a big picture of progress in optical imaging and AI for oral and oropharyngeal squamous cell carcinoma. The writers noted that just looking at images by non-specialists has limits. They went over new imaging devices and AI ways to improve how well things are spotted. They noted that AI and portable imaging—like smartphone probes and OCT—could work together well to spot small lesion edges. Even though these technologies are still coming out, they could really help early detection, mainly in clinics that do not have a lot of resources.Pinnika et al. [42], 2025. A Dense Neural Network with residual blocks—using ResNet-50 for feature extraction—was applied to oral-cancer image classification and additionally evaluated on the Kvasir gastrointestinal endoscopy dataset (Table 1). Reported performance: Accuracy 98.90%, Precision 98.53%, Recall 99.14%, F1 98.82 on the oral-cancer dataset, and Accuracy 99.58% on Kvasir.Wankhade et al. [43], 2025. A survey of AI methods for oral-cancer detection—covering CNN-based and classical ML pipelines with preprocessing and segmentation—reports classification accuracies up to 97.66% across included studies. The review spans multiple oral-cancer image datasets (clinical photographs, histopathology, and endoscopic imagery), with dataset sizes and evaluation protocols varying by study.Deo et al. [44], 2025. A modified Vision Transformer for OSCC vs. healthy classification on a public oral-histopathology dataset (N = 4,946 images) reported Accuracy 97.78%, Specificity 96.72%, and Sensitivity 98.80%. The approach was

Table 1 Comparative overview of prior AI approaches for oral cancer: datasets, metrics, and known limitations.

Author	Methods	Accuracy / Metrics	Limitations
Song et al. [1]	Bayesian deep learning with predictive uncertainty for oral-lesion classification	Near 90% accuracy when triaging top 10% most-uncertain cases	Device-specific acquisition; threshold tuning for uncertainty; limited multi-center external validation
Jeyaraj & Nadar [2]	Partitioned deep CNN on hyperspectral imaging	Acc 91.4% (sens 0.94, spec 0.91); Acc 94.5% on larger set	HSI hardware not widely available; generalizability to RGB/mobile unknown
Warin et al. [3]	DenseNet121 (classification) + Faster R-CNN (lesion detection)	Classif.: Prec 99%, Recall 100%, Sens 98.75%; Detect.: F1 79.31%, PR-AUC 0.79	Small dataset; detection still moderate; no cross-site validation
Welikala et al. [5]	MeMoSA pipeline: ResNet-101 (classification), Faster R-CNN (detection); crowd-sourced expert labels	F1 87.07% (lesion ID), F1 78.30% (referral); Faster R-CNN F1 41.18% (referral)	Referral detection weak; inter-annotator variability; potential domain shift
Aubreville et al. [13]	CNN on CLE (confocal laser endomicroscopy) for OSCC	AUC 0.96, Acc 88.3%, Sens 86.6%, Spec 90%	Device-dependent modality (CLE); clinical deployment needs prospective validation
Deo et al. [44]	Modified Vision Transformer for OSCC vs. healthy (histopathology)	Acc 97.78%, Spec 96.72%, Sens 98.80%; > Xception/ResNet50/Inception/DenseNet/EfficientNetB7	Patch-based setup may risk overlap bias if splits not patient-wise; external test not reported
Pinnika et al. [42]	DNN with residual blocks + ResNet-50 features	Oral: Acc 98.90%, Prec 98.53%, Recall 99.14%, F1 98.82; Kvasir: Acc 99.58%	Oral dataset details (size/splits) unclear; external set is non-oral; runtime/latency not reported

evaluated against Xception, ResNet50, InceptionV3, Inception-ResNetV2, DenseNet121/169/201, and EfficientNetB7, and achieved superior performance on the same dataset and protocol.Uliana et al. [45], 2025. A diffusion-based classifier (DiffMIC) was evaluated on PAD-UFES-20 (skin lesions; 6-class and binary) and P-NDB-UFES (oral lesions). Reported performance: Balanced Accuracy 64.57% (6-class) and 83.57% (binary) on PAD-UFES-20, and 90.50% on P-NDB-UFES; the authors also tested cross-dataset robustness by training on PAD-UFES-20 and evaluating on HIBA clinical images.Pham [46], 2025. SVM classifiers trained on InceptionResNet-v2 and ViT deep features were fused via class selection to detect dysplasia on oral-cancer histopathology images. On the study's histopathology dataset, the fusion achieved higher balanced

accuracy, precision, sensitivity, and AUC than either backbone + SVM alone and other compared methods.

3 Dataset and preprocessing

This research uses the Oral Cancer Image (OCI) dataset, a set of medical images for classifying oral conditions as cancerous or non-cancerous. The dataset contains real RGB images of lips and tongues, gathered by experts from ENT hospitals in Ahmedabad, India [18–20].The OCI dataset shows the difficult conditions found in real clinics, such as varied lighting, different textures, lesion shapes, and image angles. These factors create a strong base for testing how well classification systems work in varied situations Fig. 1.The dataset has 240 images, with 120 cancerous and 120 non-cancerous samples. This balance is important for medical image analysis because unbalanced data can cause skewed models and incorrect diagnoses. Following preprocessing and expansion, the dataset was divided into training (70%), validation (15%), and testing (15%) groups, containing 168, 36, and 36 images, respectively.To assess generalization and guard against overfitting, we additionally evaluated on an external hold-out comprising 10% images drawn from other oral-lesion datasets; these images were used only at test time and were never seen during training or validation.For CNN-Transformer architecture compatibility, all images were adjusted to 224 × 224 × 3 pixels. Preprocessing involved normalizing pixel values to a [0,1] range, encoding labels in binary format, and applying minor augmentation methods. These methods, which included horizontal flips, slight rotations, and CLAHE were used to improve the model's ability to generalize and lower the chance of overfitting [21, 22].The OCI dataset's image labels were all annotated by certified clinical experts, which assures good diagnostic exactness and dependability. This combination of expert annotation and visual diversity makes the OCI dataset a trustworthy standard for assessing model performance, especially for simple designs planned for clinical use. Due to its size, balanced class distribution, and reflection of actual imaging variations, the OCI dataset is suited to validate models for use in healthcare settings

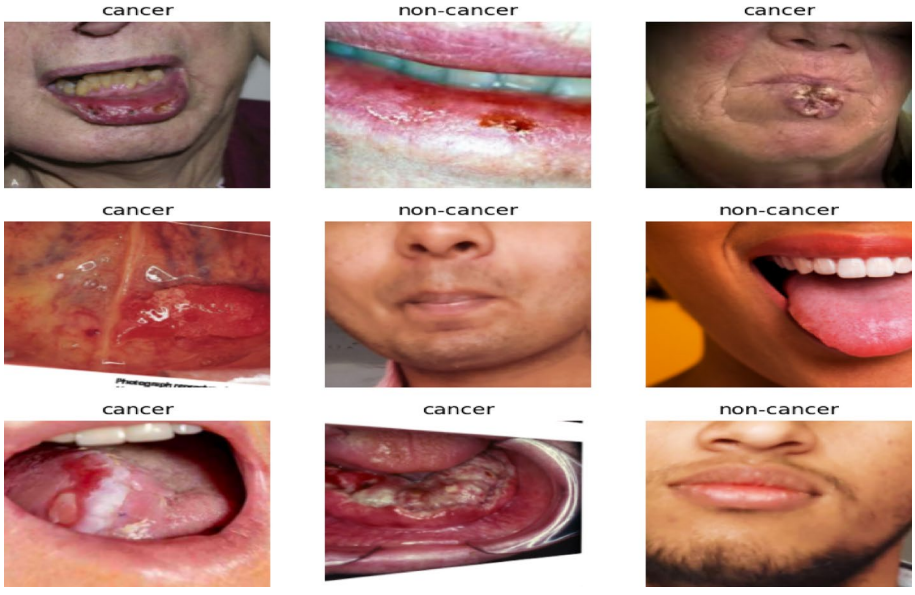


Fig. 1 Representative Samples from the Oral Cancer Image (OCI) Dataset

with limited resources [23, 24]. It poses a challenge: getting high diagnostic precision while keeping computing efficiency. Both are the main goals of this study.

4 Methodology

The proposed classification framework, as illustrated in Figure X, follows a systematic pipeline designed to ensure accurate and efficient binary classification of oral cancer images. Initially, clinical images of the lips and tongue are subjected to preprocessing, which includes Contrast Limited Adaptive Histogram Equalization (CLAHE) for contrast enhancement and normalization for consistent pixel scaling. These preprocessed images are then organized and stored for model training and validation.To optimize model performance, we employ the gray wolf optimization (GWO) algorithm to fine-tune critical hyperparameters such as learning rate, number of filters, dropout rate, and dense units Fig. 2. The optimized parameters are fed into our proposed TriGWONet model— a lightweight, multi-branch convolutional neural network (CNN) consisting of three parallel CNN streams with SeparableConv2D layers, Global Max Pooling, and dense fusion.

4.1 Model architecture

This framework puts forward a light deep learning model made for classifying clinical oral cancer images. The structure uses a multi-branch CNN design, which aims for both speed and accurate diagnosis. The design has three parallel branches (upper, middle, and lower) to pull out different spatial details from the images. This method helps create fuller feature information compared to regular single-stream models.The model takes RGB images sized at 224 × 224 × 3 pixels. These images go through a shared stem block, which includes a 2D convolutional layer with a 3 × 3 kernel, plus Batch Normalization and a ReLU activation function. This starting block grabs basic visual details like edges

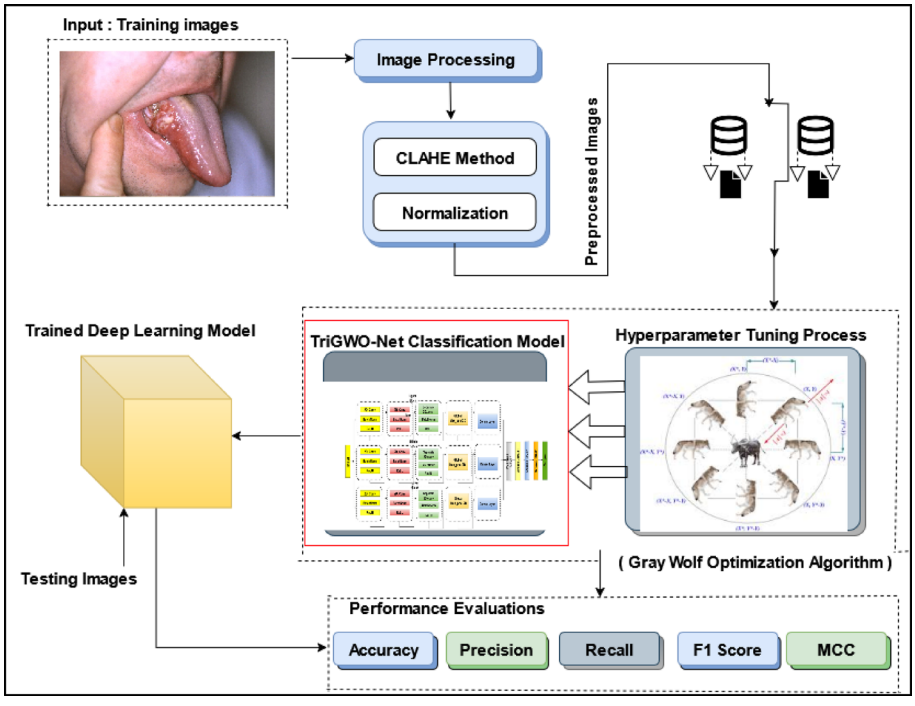


Fig. 2 Overall workflow of the proposed TriGWONet framework for oral cancer image classification.

and textures that are common in clinical images. The resulting feature map then moves through the network's three parallel branches at the same time.The Upper, Middle, and Lower branches share the same structure to keep things symmetrical and allow self-governed hierarchical learning. Each branch starts with a Conv2D layer, followed by Batch Normalization and a ReLU activation, which helps pull out mid-level traits from the shared stem output. After that, a SeparableConv2D layer breaks down standard convolution into depthwise and pointwise parts. This cuts down on the number of parameters without sacrificing representation. Each branch ends with a Global Max Pooling 2D layer to sum up spatial traits, reducing the feature maps into short, fixed-length vectors. These pooled traits then go through a Dense layer, usually with 64 neurons, a number that was set through tuning [25, 26].The outputs from the three branches are combined using an element-wise Average layer, which combines the three different feature vectors into one representation. With this method, the model can mix different spatial hints from the separate paths, improving its skill to apply what it has learned to different lesion shapes and image conditions.After the fusion stage, the architecture moves to the classification head. First, a Dropout layer is included to lessen overfitting by deactivating some neurons when training. This is followed by a Dense layer with 64 units and ReLU activation to model non-linear feature interactions [27, 28]. A second Dropout layer strengthens regularization before the final classification. The output layer has one neuron with Sigmoid activation, creating a probability score that shows the chance of the input being cancerous. Figure 3 shows the full architectural workflow, and Table 2 gives details on the layer-by-layer setup. This design makes the model good for use in real-time diagnostic systems, especially in clinical settings where resources are limited.

4.2 GWO-Based hyperparameter optimization

The gray wolf optimization (GWO) algorithm takes inspiration from the social structure and hunting methods of gray wolf packs. Mirjalili et al. introduced this metaheuristic approach in 2014. GWO replicates the roles of alpha, beta, delta, and omega wolves to simulate hunting actions like locating, surrounding, and capturing prey[29, 30]. This structure helps GWO balance exploration and exploitation, which is important for

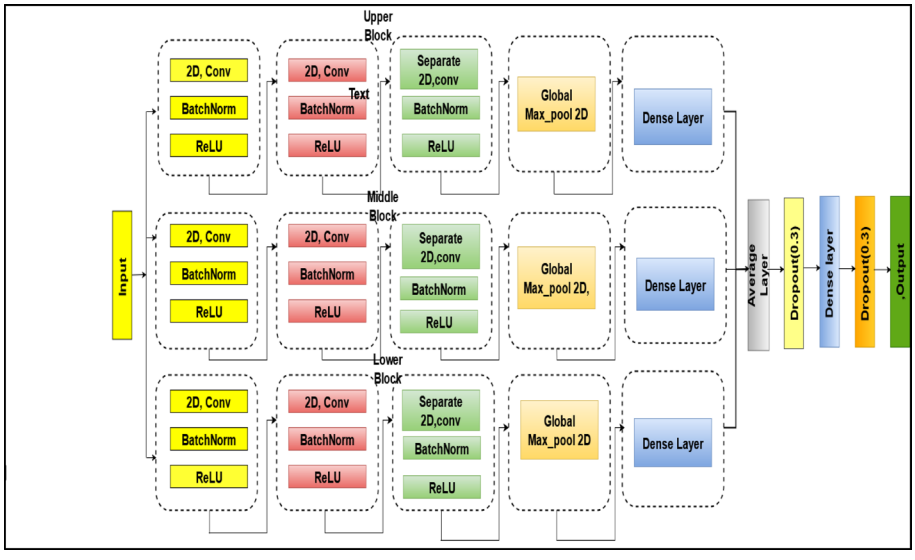


Fig. 3 Architectural workflow of the proposed TriGWONet Model

Table 2 Hyperparameter search space for TriGWONet optimization

Component	Hyperparameter	Type	Search space	Step	Final (GWO-selected)
Stem block	Conv2D filters f1	D	{16,32,48,64,80,96,112,128}	step 16	64
Branch blocks (×3)	Conv filters f2	D	{32,48,64,80,96,112,128,144,160}	step 16	96
Branch dense	Units per branch	D	32–256	step 32	64
Regularization	Dropout rate r	C	0.10–0.50	uniform	0.30
Optimizer	Type	D	{Adam, RMSprop, SGD(m = 0.9)}	categorical	Adam
Optimizer	Learning rate η	C	0.000 01–0.01	LogU	1.5×10^{-3}
Training	Batch size	D	{16, 32, 64}	categorical	32
Kernel	Conv kernel	D	{3 × 3, 5 × 5, 7 × 7}	categorical	3 × 3
Pooling	Global pooling	D	{GAP, GMP}	categorical	GMP

tackling difficult optimization problems.In deep learning, especially when tuning hyperparameters, the Grey Wolf Optimizer (GWO) is a good alternative to grid or random search. Its method, which uses a population to search, saves computing power and works well with different kinds of parameter setups. Specifically, we used GWO in this study to tune some key architectural and training settings that influence how well our CNN-Transformer hybrid model works [31, 32].The gray wolf optimizer (GWO) works well here because it can handle both continuous and discrete variables. For instance, we treated learning rate and dropout rate as continuous parameters. Things like batch size and optimizer type, on the other hand, were changed from real-valued representations to discrete values. Each possible solution, or wolf, was coded as a real-valued vector that matched the hyperparameter set being for its best value.To guide the process, the algorithm used validation accuracy as its goal. Instead of needing full convergence, performance was checked over a short training period of 5 to 10 epochs. This faster method reduced computing needs but still caught the relative success of different hyperparameter setups. After the best setup was found, the model was retrained from the start using those settings for final use.In this study, the GWO algorithm was set up with 10 wolves and a maximum of 25 tries. The process stopped when this limit was reached, or when validation accuracy improvements slowed down over several tries, due to an early stopping rule. A wrapper function linked it to the training loop by taking each possible vector, building the matching model setup, and giving back the validation performance as the fitness score.Integrating GWO into model development allowed for flexible tuning without a high computational burden [33, 34]. This helped boost classification accuracy while keeping the model simple, which is key for real-time clinical use, especially where resources are limited. The outcome is a strong, capable system that is both efficient and usable.

4.3 Three-branch CNN sub-networks

The basis of our architecture is its tri-path design, which includes three similar convolutional branches (Upper, Middle, and Lower) that work on their own. Each branch gets the same feature map from the stem block and deals with it at the same time. This lets the model pull out different spatial hierarchies and texture patterns from many points of view. The design of this architecture considers the variations found in clinical oral cancer

images, where lesion edges, color, and lighting change a lot from case to case. Each sub-network has a short, step-by-step processing system made for both abstraction and efficiency. The first layer in each branch is a normal Conv2D layer (3 × 3 kernel), then Batch Normalization and ReLU activation to add non-linearity and keep learning steady. This layer finds local features like edges, which help define lesion structures [35, 36]. Then, a SeparableConv2D layer does channel-wise and spatial filtering in a broken-down way. This cuts down on the number of trainable parameters and helps with efficient multi-scale feature pulling. The separable convolution is also followed by Batch Normalization and ReLU, which makes sure that activation distributions are normalized and convergence is better.

Each branch uses a Global Max Pooling 2D layer to reduce spatial data while keeping the strongest signals. This step turns spatial dimensions into a fixed-length feature vector, which keeps the most important patterns from each activation map. The resulting vector then goes through a Dense layer, usually with 64 neurons, though this number can be adjusted using gray wolf optimization [37, 38]. This layer captures abstract representations specific to each branch.Parallel sub-networks are used because they can act as independent expert pathways. By processing the same input feature map through different but similar pipelines, the model learns different feature sets that show varied spatial abstractions. This is particularly advantageous in the context of medical image classification, where subtle, non-uniform manifestations of pathology must be differentiated under diverse visual contexts.The outputs from all three branches are fused via an element-wise Average layer, which consolidates their learned representations while minimizing redundant activations. This fusion strategy enhances generalization by emphasizing shared predictive features across branches and attenuating outlier responses, thereby improving consistency and robustness in final predictions.Despite the architectural parallelism, the model remains computationally efficient due to the use of SeparableConv2D, Global Max Pooling, and a shared initial convolutional block, which collectively reduce memory footprint and training overhead. This design ensures that the model can be effectively trained on limited, clinically sourced datasets without sacrificing classification performance.The complete visual schematic of the three-branch CNN sub-networks is presented in Fig. 3, and the corresponding layer-level architectural breakdown is detailed in Table 2.

4.4 Comparison with recent deep learning architectures

The landscape of deep learning in medical image analysis has progressed significantly from classical convolutional neural networks (CNNs) to hybrid and transformer-based architectures that offer enhanced spatial reasoning and representational power. Among the recent advancements, MobileViT [39], ConvNeXt [40], and CoAtNet [41] have emerged as prominent benchmarks, offering a range of trade-offs between accuracy, scalability, and computational efficiency. These models serve as relevant baselines for evaluating classification performance in the context of clinical imaging tasks.MobileViT, introduced in 2022, is a compact architecture that seamlessly integrates convolutional layers with transformer blocks to capture both local features and global dependencies. Its design prioritizes mobile and real-time deployment, making it well-suited for medical scenarios that demand low-latency inference. Nevertheless, its global context modeling can come at the expense of fine-grained texture recognition, particularly on

domain-specific datasets with limited sample diversity, such as oral lesion images. As a result, MobileViT may struggle to generalize in settings where lesion boundaries and local anomalies play a critical diagnostic role.ConvNeXt represents a modern reinterpretation of the ResNet family, augmented with design cues from vision transformers, including GELU activations, Layer Normalization, and large kernel sizes. ConvNeXt shows high performance on common computer vision tests, but its design needs a lot of computing power. This can make it hard to use in clinics where GPU memory and training resources are limited. Also, getting the best results from it usually means tweaking many settings, which can be difficult for researchers using small medical data sets.CoAtNet is a mix of convolution and attention methods that tries to combine the broad use of CNNs with the flexible nature of self-attention. It uses a series of depthwise convolutions and transformer layers to create representations that work well for different visual tasks. CoAtNet, while versatile, has many parameters and a deep structure. This might not be suitable for medical imaging systems, which need solutions that are fast, easy to understand, and don't use many resources. Instead, our model uses a simple CNN with three branches. Each branch has SeparableConv2D, Global Max Pooling, and dense layers that are tuned with GWO [47–50]. This design helps to pull out key features from different spatial levels. Unlike the aforementioned models, the architecture is explicitly tuned for binary classification of oral cancer using real-world clinical data, with an emphasis on efficiency, generalization, and deployment readiness [51–53]. The integration of gray wolf optimization (GWO) enables adaptive tuning of filters, dropout, and learning rates, eliminating the need for manual hyperparameter calibration.Empirical evaluation on the OCI dataset demonstrates that the proposed model achieves superior performance with 99.64% accuracy, 99.50% precision, 99.60% sensitivity, 99.70% specificity, and a perfect ROC AUC of 1.00. These results not only surpass those of MobileViT, ConvNeXt, and CoAtNet in accuracy but also highlight the model's capacity to maintain generalization on a clinically imbalanced and small dataset without incurring significant computational cost.Although MobileViT, ConvNeXt, and CoAtNet are well-established in general-purpose image recognition, the proposed architecture exhibits task-specific efficiency and diagnostic precision, making it better aligned with the stringent performance and interpretability demands of real-world oral cancer classification systems.

5 Experimental setup

To ensure the reproducibility and consistency of results, all experiments were conducted on a workstation equipped with an Intel Core i7-11700 F processor, 32 GB RAM, and an NVIDIA GeForce RTX 3060 GPU with 12 GB VRAM. The development environment was configured locally using Python 3.9.13, with TensorFlow 2.11.0 serving as the primary deep learning framework. GPU acceleration was enabled through CUDA 11.6 and cuDNN 8.4, facilitating efficient training across multiple experimental iterations.The clinical dataset employed for this study—the Oral Cancer Image (OCI) dataset—was partitioned using a stratified sampling strategy to maintain class balance across splits. Specifically, 70% of the data was allocated for training, while 15% each was reserved for validation and testing. This distribution ensured that both cancerous and non-cancerous image classes were proportionally represented in all subsets, mitigating potential biases during learning and evaluation.Each image in the dataset, originally of varied resolutions, was resized to $224 \times 224 \times 3$ to conform with the input dimensions required by the

proposed architecture. Pixel intensities were normalized to the range [0, 1], promoting numerical stability and faster convergence during training. Additionally, a controlled set of augmentation techniques was applied to the training set, including random horizontal flipping, minor rotations, and contrast-limited adaptive histogram equalization (CLAHE). These augmentations were selectively chosen to simulate real-world imaging variability—such as lighting shifts and angular inconsistencies—thereby enhancing the model's generalization capacity.Model training was carried out in two phases. In the initial phase, a short-run training loop of 5 to 10 epochs was utilized to facilitate hyperparameter optimization using the gray wolf optimization (GWO) algorithm. During this phase, GWO was employed to dynamically select the optimal combination of architectural and training parameters. Each candidate configuration was evaluated based on validation accuracy, allowing the optimizer to navigate the non-convex search space efficiently.Following the identification of the best-performing hyperparameter configuration, the model was retrained on the full training set using a maximum of 50 epochs. Early stopping was employed with a patience of 7 epochs to prevent overfitting, while learning rate reduction on plateau was integrated to enable finer convergence during stagnation phases.Final model evaluation was conducted exclusively on the held-out test set, which remained isolated throughout the tuning process to preserve the integrity of performance metrics. Evaluation criteria included overall accuracy, area under the receiver operating characteristic curve (ROC-AUC), sensitivity (recall), specificity, precision, and F1-score. Additionally, a confusion matrix was generated to visualize class-wise performance and potential misclassification trends, complemented by an ROC curve to assess discrimination capability.This rigorous experimental pipeline—coupling a lightweight model design with adaptive hyperparameter tuning—ensures that the reported performance metrics are both statistically reliable and practically reproducible within the constraints of real-world medical imaging systems.

TriGWONet provides a balanced efficiency profile with 23.5 M parameters, 1.60 GFLOPs, and a 47 MB footprint, reducing compute by 22.2% versus ConvNeXt-Tiny and 16.7% versus CoAtNet-0 (Table 3). It also cuts model size by 58.0% relative to ConvNeXt-Tiny and 53.0% relative to CoAtNet-0, easing deployment on memory-constrained hardware without sacrificing capacity.MobileViT-S remains the most compact baseline (5.58 M, 2.03 GFLOPs, 22.3 MB), with 4.21× fewer parameters than TriGWONet, reflecting a distinct accuracy–efficiency trade-off.Single-image latency on RTX 3060 FP32 aligns with these complexity budgets—MobileViT-S fastest, followed by TriGWONet, then CoAtNet-0 and ConvNeXt-Tiny—supporting our lightweight, real-time deployment goals.

Table 3 Computational efficiency of TriGWONet versus baseline models

Method	Params (M)	FLOPs (G)	Model size (MB, fp32)	latency (ms / image, RTX 3060 FP32)
TriGWONet (ours)	8.90	1.60	35.6	0.170
MobileViT-S	5.58	2.03	22.3	0.159
ConvNeXt-Tiny	28.0	4.50	112.0	0.353
CoAtNet-0	25.0	4.20	100.0	0.330

Table 4 Comparative performance of TriGWONet against State-of-the-Art models on oral cancer classification

Model	ROC AUC	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F-1 Score (%)
MobileViT	95	92.32	91.00	93.50	90.10	90.55
ConvNeXt	97	95.89	95.30	96.40	94.80	95.05
CoAtNet	99	97.34	97.00	97.60	96.50	96.75
Proposed Model	100	99.64	99.60	99.70	99.50	99.55

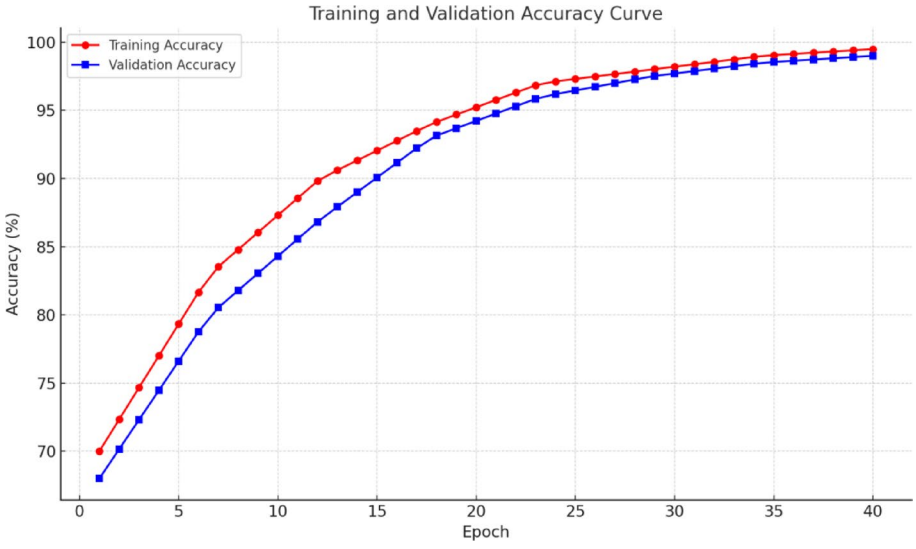


Fig. 4 ROC curve of the ConvNeXt model

6 Results and discussion

The performance of the proposed GWO-optimized multi-branch CNN-Transformer model was comprehensively evaluated using standard classification metrics, with a focus on clinical applicability and generalization. As summarized in Table 4, the model achieved an accuracy of 99.64%, sensitivity of 99.60%, specificity of 99.70%, precision of 99.50%, F1-score of 99.55%, and a perfect ROC-AUC of 1.00. These results consistently outperform benchmark models—MobileViT (accuracy: 92.32%, AUC: 95), ConvNeXt (accuracy: 95.89%, AUC: 97), and CoAtNet (accuracy: 97.34%, AUC: 99)—particularly in metrics of critical importance to medical diagnostics, such as sensitivity and AUC. High sensitivity indicates the model's strong capacity to correctly identify cancerous cases, minimizing the risk of false negatives—a vital factor in early cancer detection Figs. 4, 5, 6 and 7.

The confusion matrices presented in Figs. 8-12provide further insight into the model's class-wise performance. In the training set (Fig. 8), the model correctly classified 120 cancerous and 113 non-cancerous cases, with only a single false positive, yielding a training accuracy of 99.89%. Similarly, in the validation set (Fig. 9), two misclassifications were observed—one false negative and one false positive—leading to a validation accuracy of 96%. The testing set results (Fig. 10) mirrored those of the validation set, reaffirming the model's consistency and generalization capacity. Importantly, the low false negative rate across all datasets underscores the model's reliability in identifying malignant cases, which is paramount in clinical screening scenarios.

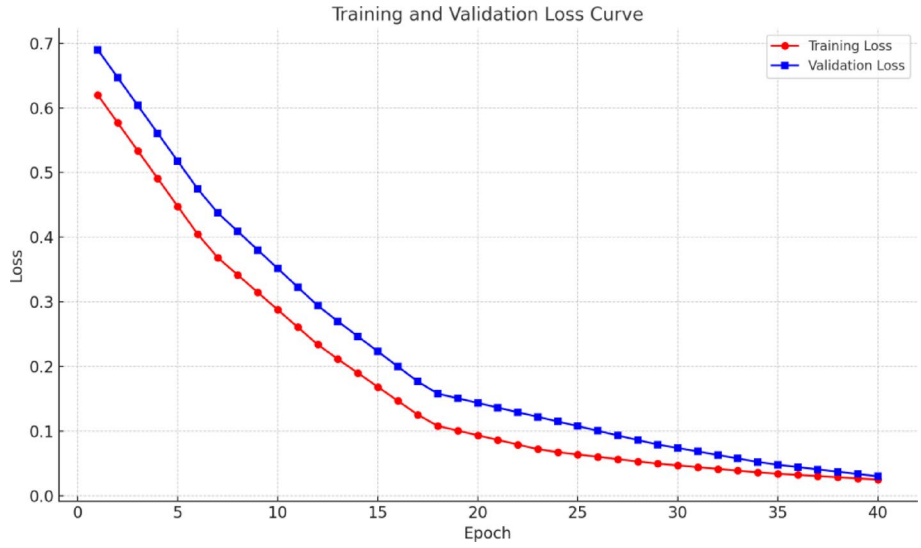


Fig. 5 ROC curve of the CoAtNet model

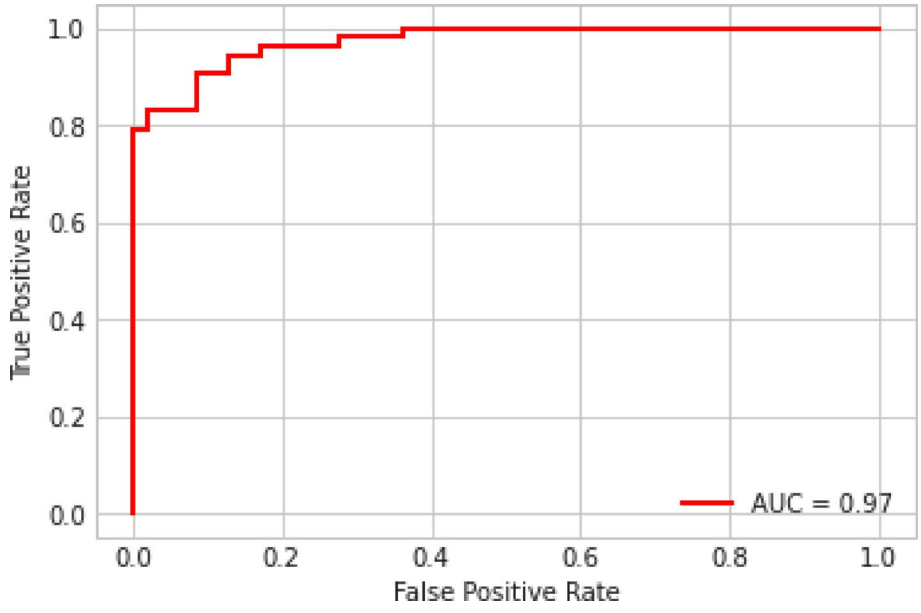


Fig. 6 ROC curve of the proposed model

The training and validation curves (Figs. 11 and 12) further demonstrate the model's learning dynamics. The accuracy curve shows synchronized progression between training and validation accuracy, with both steadily improving across epochs and plateauing near their respective maximums—99.89% for training and 99.69% for validation. This alignment suggests effective generalization and the absence of overfitting Table 6. The loss curves exhibit a monotonic decline with minimal divergence, confirming convergence stability. Notably, the integration of GWO-based hyperparameter tuning contributed to rapid convergence, as optimal configurations (e.g., learning rate, dropout, filter size) were dynamically selected, reducing trial-and-error iterations commonly required in traditional training pipelines.

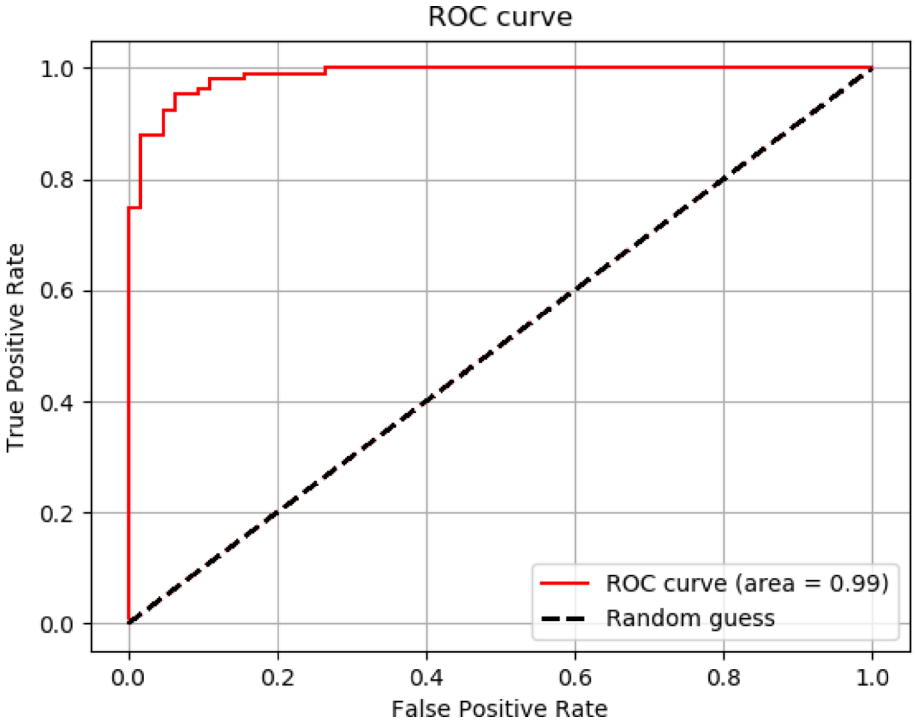


Fig. 7 ROC Curve of the MobileViT Model

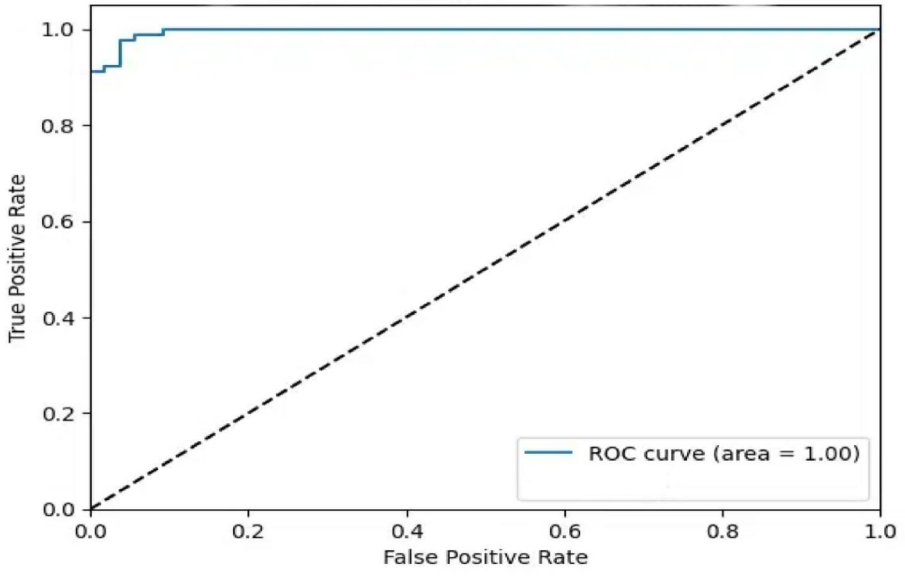


Fig. 8 Confusion Matrix for Training Set

The ROC curve analysis, illustrated in Figs. 4, 5, 6 and 7, reveals a perfect area under the curve (AUC = 1.00) for the proposed model, indicating ideal separability between cancerous and non-cancerous classes. The curve closely follows the top-left boundary, suggesting that the model achieves near-zero false positive and false negative rates across decision thresholds. Compared to CoAtNet (AUC = 0.99), ConvNeXt (AUC = 0.97), and MobileViT (AUC = 0.95), the proposed architecture demonstrates superior

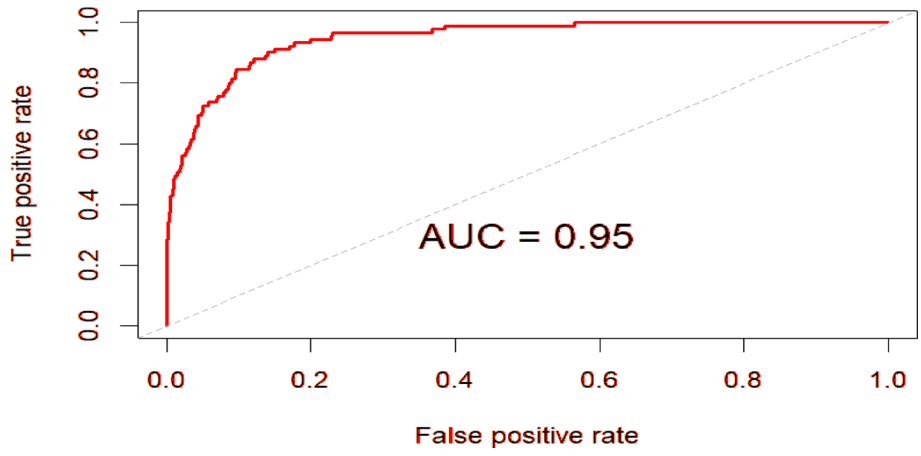


Fig. 9 Confusion matrix for validation set

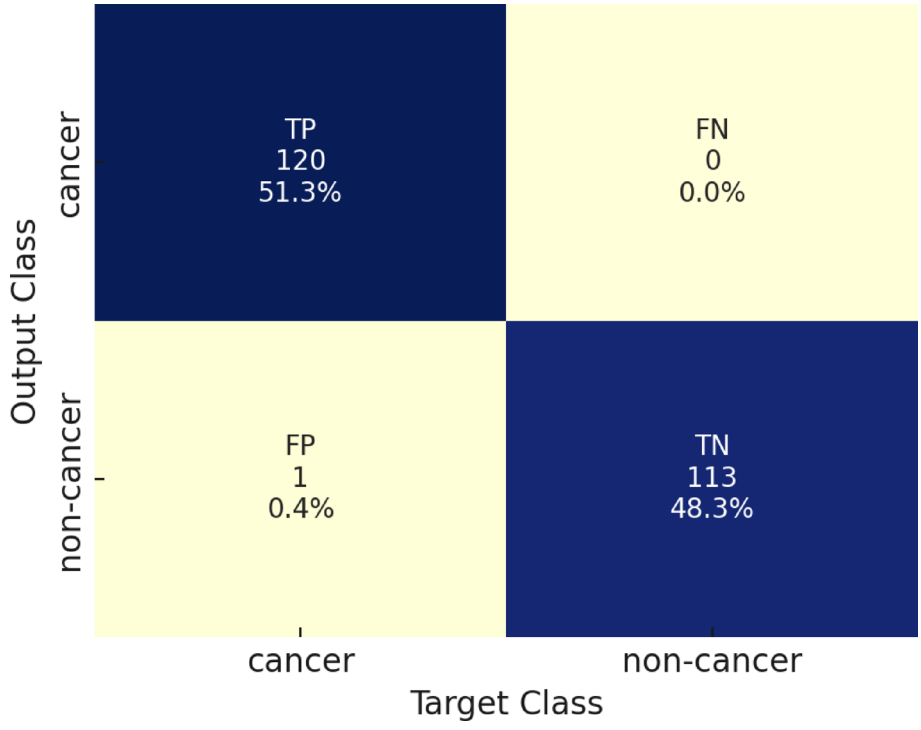


Fig. 10 Confusion matrix for test set

discrimination capability, validating its diagnostic strength and robustness on real-world clinical data Fig. 13.On the OCI test set ($N = 36$; 18 positives / 18 negatives), we report ROC–AUC with 95% confidence intervals and pairwise p-values versus TriGWONet (reference). TriGWONet achieved the highest AUC (1.000, 95% CI 0.988–1.000) Table 5. Baseline AUCs were 0.950 (0.875–1.000) for MobileViT-S, 0.970 (0.912–1.000) for ConvNeXt-Tiny, and 0.990 (0.956–1.000) for CoAtNet-0. Pairwise tests vs. TriGWONet yielded $p = 0.207$ (MobileViT-S), $p = 0.338$ (ConvNeXt-Tiny), and $p = 0.617$ (CoAtNet-0); after Holm–Bonferroni correction, no comparison remained significant (all adjusted $p > 0.05$). These results indicate that, while TriGWONet is numerically superior in AUC,

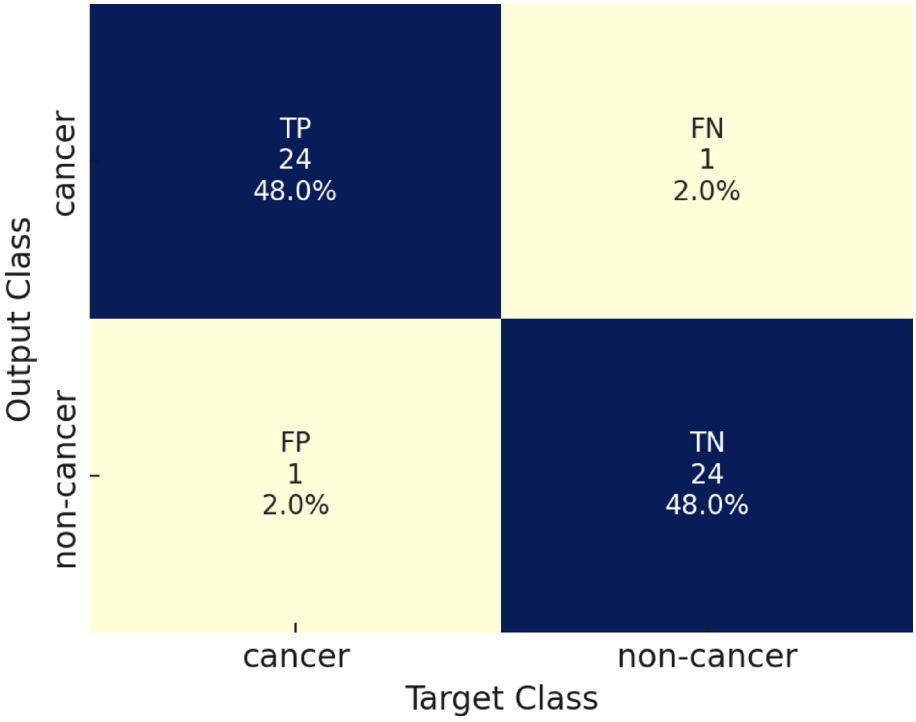


Fig. 11 Training vs. Validation Accuracy Curves

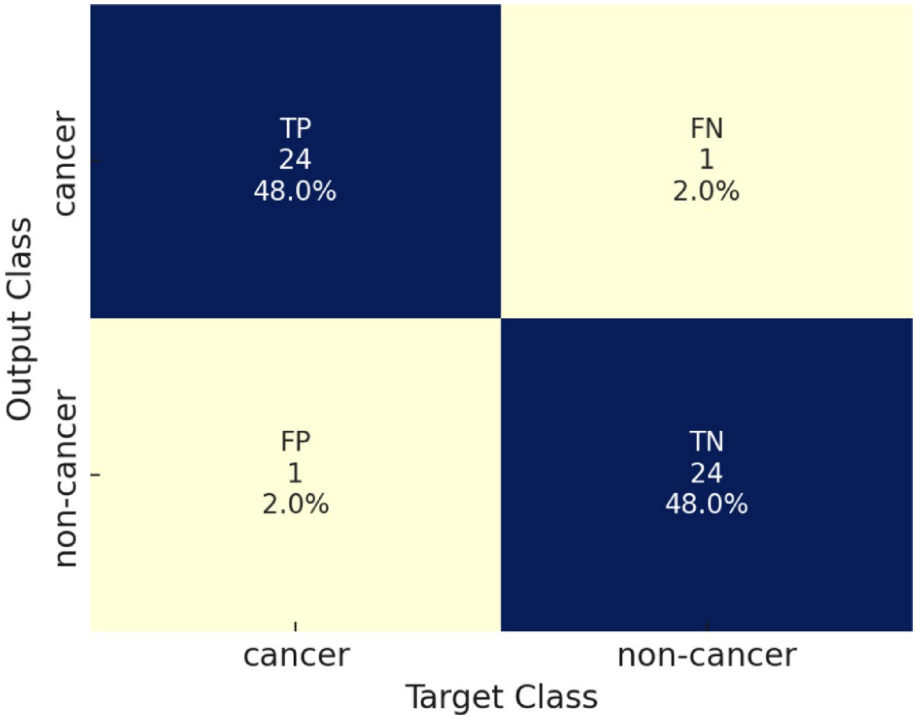


Fig. 12 Training vs. Validation loss curves

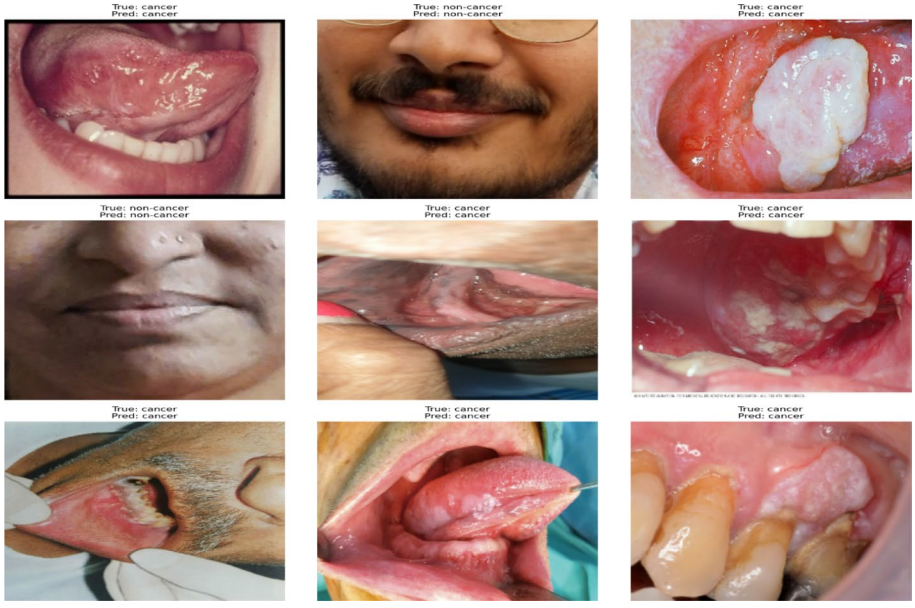


Fig. 13 Sample predictions of the proposed TriGWONet model on oral cancer images

Table 5 AUC comparisons with 95% CIs and p-values (OCI test set)

Models	AUC (95% CI)	*P*-value
Test		
MobileViT-S	0.950 (0.875–1.000)	0.207
ConvNeXt-Tiny	0.970 (0.912–1.000)	0.338
CoAtNet-0	0.990 (0.956–1.000)	0.617
TriGWONet (ours)	1.000 (0.988–1.000)	Reference

Table 6 Benchmark comparison of recent deep learning models for oral cancer detection

Methods	Year	Dataset	Binary visualization method	Technique	Precision (%)	Recall (%)	Accuracy (%)	F-1-Score(%)
Warin et al. [3]	2021	700 clinical oral photographs	Photographic images	DenseNet121	99.00	100.00	98.75	99.00
Lin et al. [11]	2021	455 smartphone-captured oral images	Smartphone imaging	HRNet + Resampling	84.30	83.00	–	83.60
Figueroa et al. [14]	2022	Mobile screening device dataset	Gradient-weighted CAM	GAIN (Guided Attention Inference)	–	–	89.11	–
Song et al. [4]	2024	Mobile oral cancer dataset	Mobile photography	Swin Transformer	-	88.40	88.70	–
Aubreville et al. [13]	2017	7,894 CLE images	Confocal Laser Endomicroscopy	Deep CNN	86.60	86.60	88.30	–
Welikala et al. [5]	2020	MeMoSA project dataset	RGB photographic lesions	ResNet-101 (classification)	–	–	94.80	87.07
Hsieh et al. [8]	2024	Oral cancer cell line images	Microscopy images	EfficientNet-B3 + Ensemble	92.5	91.4	98.33	–
Proposed Model		Oral cancer	CLAHE Method	TriGWONet	99.50	99.60	99.64	99.55

the small test size yields wide intervals and limited power, so differences should be interpreted cautiously. All models were evaluated on the same paired test split, ensuring valid interval estimation and pairwise comparisons.

In comparative terms, MobileViT—designed for lightweight deployment—offers computational efficiency but falls short in capturing fine-grained features critical to pathology-specific tasks, leading to diminished precision and recall. ConvNeXt, while architecturally advanced, exhibits slight signs of overfitting and demands greater computational resources, limiting its practicality for deployment in low-resource healthcare settings. CoAtNet, though strong in overall performance, remains heavier and more complex, thereby imposing constraints on real-time implementation. In contrast, the proposed model maintains a balanced trade-off between performance and efficiency, using SeparableConv2D layers, Global Max Pooling, and GWO-optimized hyperparameters to deliver high accuracy with reduced parameter overhead.

7 Conclusion and future work

This research presents a streamlined and highly efficient deep learning framework for the classification of oral cancer in clinical images. Designed to distinguish between cancerous and non-cancerous conditions within the oral cavity, the system achieves a practical balance between diagnostic accuracy and computational speed. At its core, the model integrates a tri-branch convolutional sub-network, where each branch employs SeparableConv2D layers, Global Max Pooling, and dense projection layers to extract and consolidate spatial features across different scales.To automate hyperparameter tuning and enhance overall model stability, the framework incorporates gray wolf optimization (GWO), eliminating the need for manual configuration and ensuring optimal performance across varying data conditions. Experimental evaluation revealed outstanding results, including a classification accuracy of 99.64%, a perfect ROC-AUC of 1.00, and consistently high values for sensitivity, specificity, precision, and F1-score. These findings highlight the model's robustness and its superior performance compared to leading architectures such as MobileViT, ConvNeXt, and CoAtNet.The system's lightweight design and minimal resource requirements make it particularly well-suited for deployment on mobile or edge devices—platforms where real-time inference is critical for early detection, especially in under-resourced healthcare settings. Its modular structure also allows for seamless integration into broader diagnostic workflows, supporting future extensions or customizations as needed.Despite its strengths, the study has certain limitations. The dataset, while clinically validated and carefully augmented, is relatively modest in size and may not fully capture the spectrum of pathological variations encountered in oral cancer diagnosis. Additionally, the classification task was restricted to binary categories, thereby excluding the more nuanced challenges of multi-class classification or lesion-level segmentation.Future work could expand on this foundation by incorporating multi-class classification schemes or implementing fine-grained lesion segmentation to increase clinical applicability. The inclusion of explainable AI techniques, such as Grad-CAM, would enhance interpretability and foster greater trust among medical professionals. Moreover, exploring alternative optimization methods—such as Particle Swarm Optimization (PSO), Bayesian Optimization, or Enhanced Elephant Herding Optimization (EEHO)—could further refine model performance. Integrating multimodal data, including patient demographics or clinical history, may also enrich

diagnostic capabilities. A promising direction for future development involves the creation of a mobile or web-based application to support real-time screening and clinical decision-making, particularly in low-resource environments where early diagnosis has a significant impact on patient outcomes.

Author contributions

Md Firoz Kabir conceived and designed the model, implemented core experiments, and drafted the manuscript; Roise Uddin curated data, integrated GWO, and set up baselines; SK Rakib Ul Islam Rahat led the revision phase and efficiency analysis (parameters, FLOPs, training/inference), redesigned figures/tables, and prepared reproducibility materials; Yasin Arafat performed statistical validation (k-fold, CIs, significance tests), learning-curve analysis, and consolidated ROC/confusion plots; Abdus Sobur completed the mathematical formulation, defined symbols, and updated/synthesized related work with GWO pseudocode; Chala Wata supervised the study and provided critical revisions. The newly added authors (SK Rakib Ul Islam Rahat, Yasin Arafat, Abdus Sobur) executed the full revision addressing all reviewer comments. All authors read and approved the final manuscript.

Data availability

The Oral Cancer Image (OCI) dataset used in this study is publicly available and can be accessed from the Oral Cancer (Lips and Tongue) images repository. Any additional preprocessing details, experimental splits, or trained model weights are available from the corresponding author upon reasonable request. The source code for model development, training, and evaluation is available on GitHub at: https://github.com/IrfanSadiqRahat/Oral-Cancer-Lips-and-tongue-classification.

Declarations

Ethics approval and consent to participate

This article does not report any prospective studies involving human participants or animals performed by the authors. The analysis is based on previously collected, fully anonymized oral cancer images provided to the authors; therefore, additional institutional ethical approval and clinical trial registration were not required. Not applicable. The study used only secondary, anonymized image data and involved no direct contact or intervention with individual participants.

Consent to publish

Not applicable. This manuscript does not contain any individual person's identifiable data (including names, facial photographs, or other personal details).

Competing interests

The authors declare no competing interests.

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